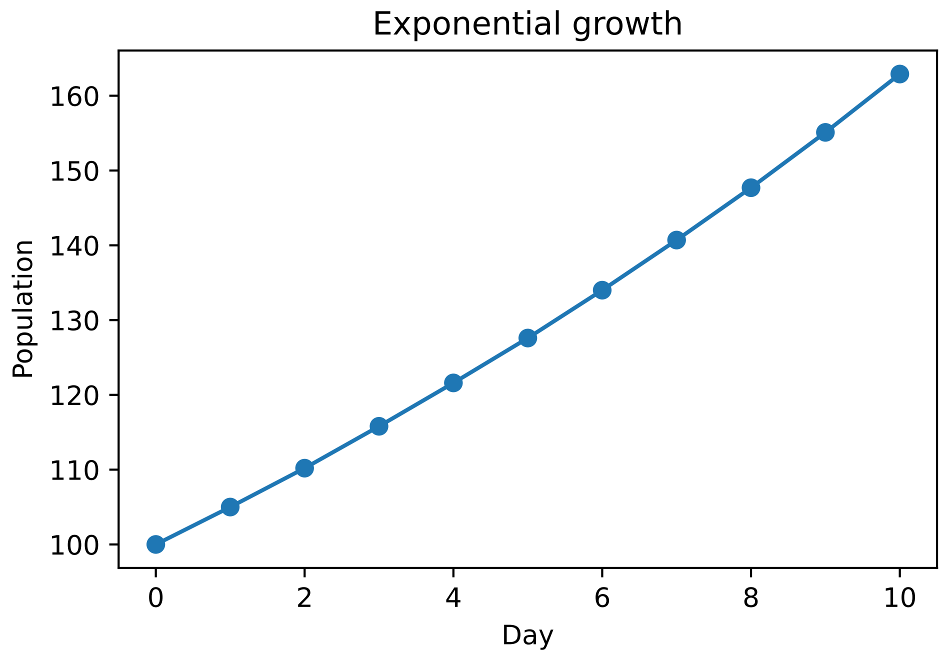
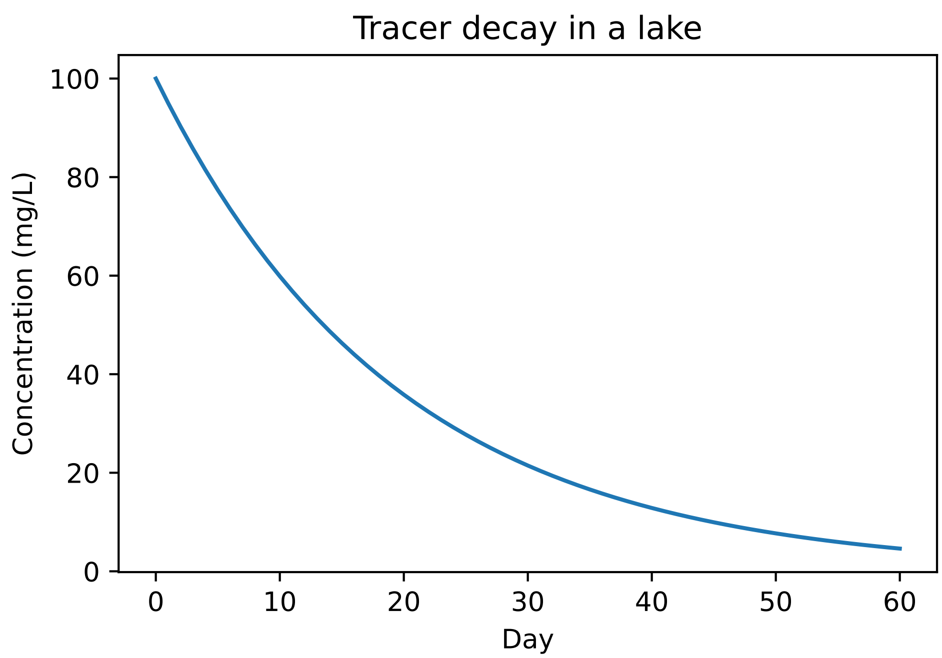
Lab 1 warm-up

```python
print("Hello World")

# Calculate the volume of a rectangular estuary (length x width x depth)
length = 5000   # metres
width = 800     # metres
depth = 3.5     # metres
volume = length * width * depth
print(f"Estuary volume: {volume} m^3")

# Simulate 10 days of exponential population growth
population = 100
rate = 0.05
history = [population]

for day in range(10):
    population = population + rate * population
    history.append(round(population, 1))

print(history)

# Write a function that converts temperature from Celsius to Fahrenheit
def celsius_to_fahrenheit(c):
    return c * 9/5 + 32

# Test it
for temp in [0, 15, 25, 37]:
    print(f"{temp}°C = {celsius_to_fahrenheit(temp)}°F")

    import matplotlib.pyplot as plt

days = list(range(len(history)))
plt.plot(days, history, "o-")
```

```python
plt.xlabel("Day")
plt.ylabel("Population")
plt.title("Exponential growth")
plt.show()

import matplotlib.pyplot as plt

concentration = 100.0
decay_rate = 0.05
dt = 1.0
history = [concentration]

for day in range(60):
    concentration = concentration - decay_rate * concentration * dt
    history.append(concentration)

plt.plot(history)
plt.xlabel("Day")
plt.ylabel("Concentration (mg/L)")
plt.title("Tracer decay in a lake")
plt.show()
```

```
Hello World
Estuary volume: 14000000.0 m^3
[100, 105.0, 110.2, 115.8, 121.6, 127.6, 134.0, 140.7, 147.7, 155.1, 162.9]
0°C = 32.0°F
15°C = 59.0°F
25°C = 77.0°F
37°C = 98.6°F
```


Exponential growth
Population
Day
100
110
120
130
140
150
160
0
2
4
6
8
10


Tracer decay in a lake
Concentration (mg/L)
Day
0
20
40
60
80
100
0
10
20
30
40
50
60